

Erdős Problem #263

EP-263

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let a_n be a sequence of positive integers such that for every sequence of positive integers b_n with $b_n/a_n \rightarrow 1$ the sum $\sum \frac{1}{b_n}$ is irrational. Is $a_n = 2^{2^n}$ such a sequence? Must such a sequence satisfy $a_n^{1/n} \rightarrow \infty$?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>